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Development of an AD5933-based impedance calibration and measurement technology using piezoceramic transducers

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A R T I C L E I N F O

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A B S T R A C T

AD5933 evaluation board is expected to promote the practicability of electromechanical impedance (EMI) based structural damage identification technology. For the evaluation board, the calibration before use is essential and important. In this study, a novel self-calibration method is proposed to help the board to improve its independence, measurement accuracy, and data continuity for the impedance measurement by selecting the optimal feedback resistance based on data deviation characteristics of the AD5933 board. In this research, to clearly present the principle of the proposed method, a pretest of impedance measurement via AD5933 board was conducted at first. Then, its impedance measurement performance was further investigated by a series of measurements tests, and the results demonstrated that the proposed method provide the board the ability to obtain the impedance measurement with higher accuracy and better data continuity comparing with the conventional subrange calibration method. At last, its feasibility and effectiveness for structural damage identification has also been validated by a bolt loosening identification test on a steel plate.

1. Introduction

Benefiting from the distinct advantages, such as being sensitive to the tiny structural defects [1–4], the potential to be applied in complex structures [5,6] and so on, the electromechanical impedance (EMI)-based structural damage detection approach has received lots of attentions, and been regarded as one of the most promising NDT technologies [7–11]. In recent decades, with the help of professional impedance measurement instrument, the EMI technology has been employed to detect various kinds of structural defects and damages, such as the bond slip between steel profiles and concrete [12], the debonding in FRP-strengthened reinforced concrete beams [13] and honeycomb sandwich composite materials [14], the pipe looseness [15], the crack in the pipeline [16] and so on. However, it still faces a great challenge for the EMI-based damage detection technology to be applied in practical applications. One of the main limitations is that the professional impedance measurement instrument is usually very expensive, bulky and inefficient [17]. Therefore, it is always difficult to be applied for practical engineering structures, and finally seriously limiting the application and development of EMI-based method in the area of structural health monitoring (SHM).

AD5933 evaluation board (Analog Devices Inc., Norwood, MA, USA) as one of the miniature impedance measurement devices with the characteristics of high integration and cost-effective, makes it possible to address the huge and high-cost problems of commercialized impedance instruments during the structural health monitoring and defect identification. Benefit from these advantages, this board combining with the impedance measurement technology had received lots of attentions in the areas of SHM and NDT, for example, to detect the debonding and defect of the glass–epoxy composite plate [18], the strength development process of the concrete [19], the corrosion of the aluminum alloy beam [20], the notch damage of the concrete structure [21], the looseness of the bolt connection [22,23], the delamination and crack of the glass fiber composite plates [24] and so on. Unfortunately, although the AD5933 evaluation board and technology has been successfully applied in various kinds of structures in the laboratory, it is still difficult to be utilized independently, because the calibration process of the

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Nomenclature

List of abbreviations

ADC	analog to digital converter
CCD	cross-correlation distance
DFT	discrete Fourier transform
DSP	digital signal processing engine
EMI	electromechanical impedance
LPF	low pass filter
MRE	mean relative error
NDT	non-destructive testing
PGA	programmable gain amplifier
PZT	piezoelectric transducer
RADR	relative amplitude deviation rate
RSDR	relative shape deviation rate
RMSD	root mean square deviation
SHM	structural health monitoring

List of symbols

B	imaginary part of admittance
CCD_j	cross-correlation distance of the impedance signals corresponding to the jth and $(j+1)$th resistance
GF	gain factor
G	real part of admittance
I	imaginary part of impedance output signal for AD5933
I_1	imaginary part of the impedance for the calibration resistor
I_2	imaginary part of the impedance for the to-be-measured structure
M	modulus of the of impedance signal
M_1	modulus of the calibration resistor
M_2	modulus of the to-be-measured object
P_i^S	the ranking score of the signals corresponding to the ith feedback resistance, $S = Z, \theta, R, X, G, B$
$\overline{P_i}$	the total score of the resistance in position i
P_i^{RADR}	the score of the RADR for the ith resistance
P_i^{RSDR}	the score of the RSDR for the ith resistance
R_{CAL}	resistance of calibration resistor
R_{FB}	resistance of feedback resistor
R	real part of impedance
R_1	real part of the impedance for the calibration resistor
R_2	real part of the impedance for the to-be-measured structure
$RADR_j$	relative amplitude deviation rate corresponding to the jth resistance
$RSDR_j$	relative shape deviation rate corresponding to the jth resistance
r_{j+1}	the resistance at position $j+1$ after ranking
S_i^{rj}	the signal at the ith frequency point corresponding to the jth resistor
$\overline{S^{rj}}$	mean of the signal corresponding to the jth feedback resistances r_j
V_{in}	Input excitation voltage
V_{out}	output voltage
X	imaginary part of impedance
Z	Impedance
Z_{CAL}	calibration resistor
Z_x	impedance of the to-be-measured object
θ	phase angle after correction
θ_{system}	phase of the system of AD5933 in the calibration stage
θ_2	phase angle of the to-be-measured object in the measurement stage
ω^s	weight coefficient for the corresponding impedance parameters, $S = Z, \theta, R, X, G, B$
ω_{RSDR}	weight coefficient of RSDR value
ω_{RADR}	weight coefficient of RADR value
σ^{rj}	standard deviation of the signal corresponding to the jth feedback resistances r_j

board before use can not yet get rid of the help of the professional impedance measurement instruments [25,26]. Besides, several critical processes during the calibration, such as selecting the feedback resistance, still heavily rely on the operators' experiences, and thus seriously limit the stability and efficiency of the impedance measurement when using this board and technology.

For the AD5933 board, the calibration before use is essential and important, and usually the calibration and to-be-measured impedances must be close enough to each other for better measurement accuracy. In this situation, selecting the appropriate calibration parameters, such as the feedback resistance, is critical for the following impedance measurement. Therefore, to improve the independence of the board and make it more convenient and automatic in operation, especially for the feedback resistance selection, some attempts have been conducted by the researchers. For example, to select the appropriate feedback resistor, Guo *et al.* [27] divided the entire impedance measurement ranges of the board from 100 Ω to 10MΩ into six subranges, and each subrange corresponded to a special feedback resistor. At the beginning, the feedback resistor corresponding to the first subrange was selected and attended to the impedance measurement and calculation, then the proposed system checked whether the calculated result was within the range required by the resistor group of the first subrange. If not, then the system changed the group and started a new cycle for impedance calculation until the results succeeded to meet the requirement. Thus, its corresponding resistor of the group was regarded as the final selected feedback resistance. This kind of selection method called the subrange calibration method will be introduced in detail in the following section. Besides, another approach was also presented and utilized by Wandowski [26] to select the feedback resistor (R_{FB}) following the equation, *i.e.*, $R_{FB} = (Z_{min} + Z_{max})/3$, in which, the Z_{min}, Z_{max} denote the minimal and maximal impedance of the measured element, and can be obtained by the professional impedance analyzer at the beginning of the measurement. For this kind of method, the feedback resistor selection is easy to operate, but it still can not get rid of the professional instruments completely, besides, this approximate calculation will inevitably lead to the more serious measurement error. On the other hand, the artificial neural network-based signal post-processing algorithm was also reported by Istanbullu *et al.* [17] to deal with the calibration requirements of the AD5933 integrated circuit. In the research, an application specific artificial neural network topology was developed and trained for high-precision impedance measurement using a fixed calibration impedance. For this algorithm, the network training will cost huge workload, and it is still very difficult to collect enough training samples to cover the different frequency and impedance as much as possible. In recent years, the piecewise linear interpolation method (Lin *et al.* [28]) and the piecewise linear fitting method (Wen *et al.* [29]) were also proposed to realize the nonlinear compensation to obtain the calibration parameters including the gain factor and the feedback resistance. However, for this kind of methods, lots of the interpolation or fitting parameters need to be calculated and stored in advance, besides, not only the piecewise density and numbers but the fitting quality will also directly influence the impedance measurement accuracy.

Among these methods mentioned above, the subrange calibration method is widely used owing to its simple principle and convenient operation. Based on this calibration method, the AD5933 board has been successfully employed to monitor the impact and damage of a sailplane

[27], to measure the water conductivity [30] and so on. However, the record overflow limitation is intrinsically inevitable, which may induce the discontinuity problem for the impedance measurement especially when across the frequency subranges. In this situation, the discontinuity may induce the key information missing for the structures, thus confuse even mislead the final evaluation for the structural health condition and service status.

Therefore, in order to improve the data continuity of the impedance measurement of the board and reduce the reliance on the professional impedance measurement instruments and the manual experience, a novel self-calibration method is proposed and investigated in this paper based on the deviation characteristics of the impedance measurement by the AD5933 board. In this study, the impedance measurement principle of the AD5933 board and the procedure of the conventional subrange calibration method were firstly introduced and presented briefly. Then, the calibration principle and procedure of the proposed self-calibration method were presented in detail. After that, a series of impedance measurement tests were conducted to investigate and compare the measurement accuracy of the proposed method especially with the ones measured with the help of conventional subrange calibration method. In addition, its feasibility and effectiveness of the proposed method for the structural defect identification have also been validated by a bolt loosening investigation.

2. Methodology for the AD5933 board

2.1. Impedance measurement principle

The AD5933 evaluation board (as shown in Fig. 1) developed by Analog Devices Co. (Norwood, MA, United States), as a miniature impedance measurement device, can be employed and utilized to measure the electrical impedance of the structures with the frequency up to 100 kHz. The detailed description about this device can be referred to the literature [25]. However, it should be noted that for the AD5933 board, the appropriate calibration before use is essential and important, and the calibration result directly influences the impedance measurement performance.

For the AD5933 board, the principle and procedure of the impedance measurement can be described by the diagram in Fig. 2, in which V_{in} is the input voltage signal generated by the internal frequency generator of AD5933 with the controllable amplitude, Z_x and R_{FB} denote the to-be-measured resistance and the feedback resistance, respectively, and V_{out} is the output voltage signal.

The frequency generator of the board allows an external complex voltage signal to be excited with a known frequency, therefore the impedance magnitude $Z_x(f_i)$ at the corresponding excitation frequency f_i can be calculated and obtained according to equation (1):

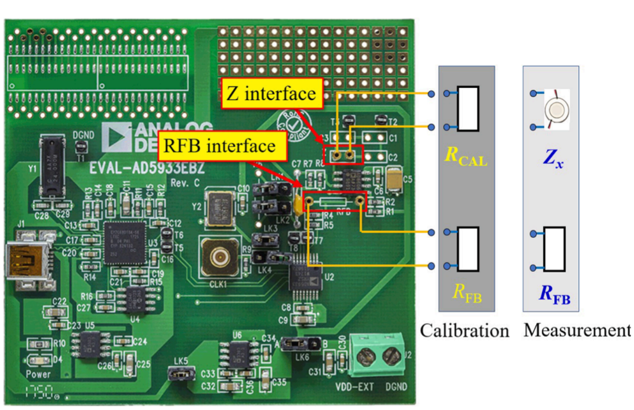


Fig. 1. AD5933 evaluation board and its layout during the calibration and measurement process.

$$Z_x(f_i) = -\frac{V_{in}(f_i)}{V_{out}(f_i)} \times R_{FB} \tag{1}$$

However, for the practical impedance measurement, V_{out} will be firstly amplified by the programmable gain amplification (PGA) and filtered by the lowpass filtering (LPF), and then transmitted to the analog-to-digital converter (ADC). Finally, following the discrete Fourier transform (DFT) operation by the digital signal processing engine (DSP) at each excitation frequency, the real (R) and imaginary (I) part of the impedance signal at each output frequency can be obtained [31].

Based on above analysis, it is found that to guarantee the impedance measurement with enough accuracy, the V_{out} must be controlled in the permissible range of ADC input after the PGA and LPF operation. Following the equation (1), it is clear that when V_{in} has been determined in advance, to meet the requirement of the V_{out}, the feedback resistance R_{FB} must be carefully selected depending on the impedance Z_x of the to-be-measured structure. In summary, for AD5933 board, the parameters of the system gain, including the resistor R_{FB} and the value of PGA, are very important during the impedance measurement.

2.2. Calibration and measurement procedure

In practical applications of the AD5933 evaluation board, once the system gain parameters including R_{FB}, V_{in} and the value of PGA, are determined, the calibration of the board and the impedance measurement of the to-be-measured structure can be conducted. In this section, the procedures of the board calibration and the impedance measurement were briefly introduced as following:

(1) Calibration of the board.

The board calibration is actually based on the calculation of gain factor (GF) for the case that the calibration impedance Z_{CAL} has been known (in this study, the pure resistor is used, thus $Z_{\text{CAL}} = R_{\text{CAL}}$) [26,32]. The R_{CAL} selection is directly related to the impedance of the to-be-measured structure and is very important during the calibration, but nowadays it still seriously depends on the professional impedance analyzer. The detailed selection procedure can be referred to the literature [26].

After the calibration resistance R_{CAL} is determined, the calibration work of the board can be done. At first, as shown in Fig. 1 for the calibration stage, the selected calibration resistor R_{CAL} was connected to the Z interface, and the feedback resistor R_{FB} (selection of $R_{\text{FB}} = R_{\text{CAL}}$) was connected to the RFB interface. Then, the related preset parameters of the board were evaluated and determined, including the PGA and V_{in}. Subsequently, the impedance measurement for the calibration resistor Z_{CAL} was conducted by the board, thus the original real and imaginary parts (R_1 and I_1) of the impedance for the calibration resistor R_{CAL} can be respectively obtained from the register on the board. Finally, the modulus M_1 of R_{CAL} in the calibration stage, the gain factor GF and the system phase angle θ_{system} can be calculated respectively according to equations (2)–(4):

$$M_1 = \sqrt{R_1^2 + I_1^2} \tag{2}$$

$$\text{GF} = \frac{1}{R_{CAL} M_1} \tag{3}$$

$$\theta_{system} = \arctan\left(\frac{I_1}{R_1}\right) \times \frac{180^{\circ}}{\pi} \tag{4}$$

(2) Impedance measurement of the structure.

Once the board calibration is finished, and the corresponding calibration parameters including the gain factor GF and the system phase

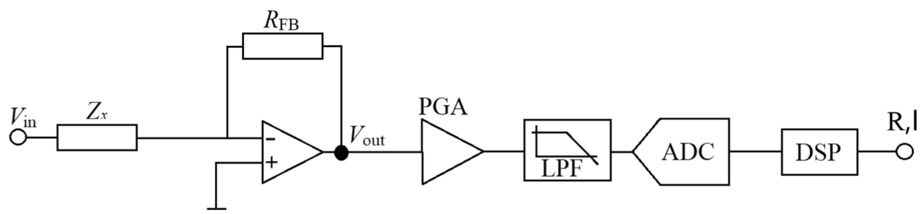


Fig. 2. The diagram of the impedance measurement using the AD5933 chip.

angel θ_{system} are obtained, then the impedance measurement for the to-be-measured structure can be conducted. At first, as shown in Fig. 1, for the measurement stage, the to-be-measured object Z_x (such as PZT) was connected to the Z interface, and the feedback resistor R_{FB} was still connected to the RFB interface. Following the impedance measurement procedure of the board, then the original real and imaginary part (R_2 and I_2) of the impedance for the to-be-measured structure under this situation can be obtained. Finally, the modulus M_2, the impedance Z_x and the phase angle θ_2 of the to-be-measured object in the measurement stage can be calculated following equations (5)–(7):

$$M_2 = \sqrt{R_2^2 + I_2^2} \tag{5}$$

$$Z_x = \frac{\frac{1}{GF}}{M_2} = \frac{M_1 \times R_{FB}}{M_2} \tag{6}$$

$$\theta_2 = \arctan\left(\frac{I_2}{R_2}\right) \times \frac{180^{\circ}}{\pi} \tag{7}$$

Similar to the calculation phase in equation (4), the phase in equation (7) should also be updated based on its given quadrant [32]. In addition, considering the influence of the system phase of the board in the calibration stage, the phase of the to-be-measured subject can be finally modified following equation (8):

$$\theta = \theta_2 - \theta_{system} \tag{8}$$

After the calibration and measurement operation mentioned above, the impedance measurement, such as the modulus $|Z|$, phase θ and so on, of the to-be-measured object can be obtained based on above analysis [25,26].

However, it should be noted that the selection of the feedback resistance R_{FB} in the calibration stage is still heavily dependent on the professional impedance measurement instruments, therefore nowadays it is difficult for the AD5933 board to be used independently. Besides, the manual experience also obviously influences the selection of the feedback resistance R_{FB}. All these issues seriously limit the development and application of the AD5933 EMI-based technology in the area of structural health monitoring, especially for the practical applications.

2.3. Subrange calibration method

In order to improve the independence and automation level of the AD5933 board during its calibration and impedance measurement, several attempts have been conducted by the researchers, such as proposing the special algorithms or methods embedded into the board to help to realize the automatic selection for the feedback resistance R_{FB}. Among these methods, the subrange calibration method is widely utilized owing to its simplicity and convenience [30].

For this kind of calibration method, the feedback resistance of the board is selected referring to the ranges of the impedance modulus M_2 of the to-be-measured subject according to equation (9), which can be derived from equations (1) and (6):

$$M_2 = -\frac{V_{out}}{GF \times R_{FB} \times V_{in}} \tag{9}$$

Considering the V_{out} in the equation (9) will be firstly amplified by the programmable gain amplification (PGA) of the board according to the description in section 2.1, so the actual output voltage, denoted by V^*_{out}, can be calculated by multiplying the value of PGA, *i.e.*, $V^*_{out} = V_{out} \times$ PGA. Thus, equation (9) can be transferred to the following:

$$M_2 = -\frac{V^*_{out}}{GF \times R_{FB} \times PGA \times V_{in}} \tag{10}$$

Based on above equation, it is clear that when V_{in} and PGA of the board are set to a fixed value, to guarantee the V^*_{out} within the linear input allowance range of ADC of the board, the feedback resistance R_{FB} and the gain factor GF should be carefully selected with different values corresponding to different impedance modulus M_2 of the to-be-measured subject.

For this subrange calibration method, the entire impedance measurement range from several ohms to megohm is divided into different subranges, as shown in Table 1, and the corresponding gain factors and feedback resistances under each subrange are calculated respectively and then saved in the board for the following impedance measurement. In addition, the feedback resistance R^i_{FB} corresponding to ith subrange is pre-selected in the range of [Z_{i-1}, Z_i]. In this situation, the corresponding gain factor GF_i for this subrange can be obtained by calculating the average of the gain factors of the feedback resistance R^i_{FB} under a fixed frequency point (such as 10 kHz) following the impedance measurement procedure in section 2.2.

However, for the specific gain factor, it can be just applied in very narrow ranges for the impedance magnitude and frequency band, and the gain factor corresponding to different impedance magnitude or sampling frequency should be calculated and calibrated separately. Unfortunately, for the subrange calibration method described in this section, the gain factor GF in each subrange is regarded as constant, which may lead to limited accuracy for the impedance measurement.

On the other hand, for this calibration method, although the subrange division of the impedance magnitude range from 0 O to megohm is continuous, the frequency blank band still exists in the impedance curve due to the inappropriate feedback resistance selection caused by the different gain factors (GF) in each subrange. As a result, the impedance modulus curve present discontinuity, as shown in Fig. 3, which will seriously affect the reliability and feasibility of AD5933 board in the area of impedance measurement and EMI-based structural damage identification. This discontinuity phenomenon will also be presented and analyzed in detail in the following experiments in section 5.1.

Table 1
Subrange calibration method.

Impedance measurement range	Feedback resistance	Gain factor
0 to Z_1	R^1_{FB}	GF_1
Z_1 to Z_2	R^2_{FB}	GF_2
⋮	⋮	⋮
Z_{i-1} to Z_i	R^i_{FB}	GF_i
⋮	⋮	⋮
Z_{m-1} to 10 MΩ	R^m_{FB}	GF_m

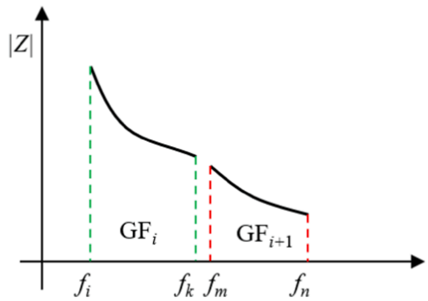


Fig. 3. The discontinuity of the impedance measurement using the AD5933 board with the help of the subrange calibration method.

3. Methodology for the proposed self-calibration method

3.1. Method principle

In this paper, in order to overcome the discontinuity issue caused by the subrange calibration method and improve the impedance measurement accuracy and independence, a novel self-calibration method was proposed to work with the AD5933 board. With the help of this method, the AD5933 board can realize the calibration operation by itself and do not need to rely on the external instrument during the impedance measurement. In addition, the continuous impedance data can be obtained by combining the AD5933 board with a fixed feedback resistance which was selected with the help of the proposed method.

As the description of equation (10) in Section 2.3, when the feedback resistance R_{FB} is selected too large or too small, the system gain factor GF will make V_{out} out of the linear allowance range of ADC, and finally induce an inaccurate result for the impedance measurement. Therefore, it is the key to search and select the optimal feedback resistance which has the potential to make the system gain most suitable and finally to help to obtain more accurate measurement result than using other feedback resistances.

On the other hand, lots of impedance measurement tests using the AD5933 board present that the system gain of the board will obviously influence the impedance measurement, and inappropriate selection may induce the deviation of the measured impedance signal, especially when comparing with the one obtained using the optimal feedback resistance. At the same time, based on the test results, a phenomenon is also observed that for the selected feedback resistance, the closer to the optimal one, the smaller of the deviation presents. This phenomenon will be presented and analyzed in detail by the pretest investigation in section 4. Based on this phenomenon combining with the measurement characteristics of AD5933, a novel self-calibration method used for AD5933 was proposed in this paper to help to select the optimal feedback resistance.

For the proposed self-calibration method, several resistors with different resistance values were prepared as the alternative feedback resistance at the beginning. Then, the alternative resistor with the smallest resistance was employed firstly as the feedback resistance to work with the AD5933 board to obtain the impedance measurement in this situation. Subsequently, the alternative resistor with the second smallest resistance was selected following the same procedure to use to obtain the corresponding impedance measurement. The similar operation was conducted for each alternative feedback resistance one by one from the smallest to the largest, and finally all the impedance data corresponding to each feedback resistance can be obtained successfully.

After that, the impedance signals measured corresponding to the adjacent feedback resistances were compared with each other, and the deviation between these signals was calculated. In this study, different kinds of impedance parameters, including the modulus $|Z|$, the phase θ, the real part R and imaginary part X of impedance, the real part G and imaginary part B of admittance (inverse of impedance), were employed for the comparation and deviation calculation. In addition, the relative amplitude deviation rate (RADR) and the relative shape deviation rate (RSDR) calculated from the cross-correlation distance (CCD) [33] were utilized to quantificationally evaluate the deviation following equations (11)–(13).

For example, the *RADR* corresponding to the jth resistance, denoted by $RADR_j$, can be calculated by the following equation:

$$RADR_j = \frac{\sum_{i=1}^{m} \left| S_i^{r_{j+1}} - S_i^{r_j} \right|}{\Delta R_j} \tag{11}$$

where r_j denotes the jth resistor when the alternative feedback resistances have been arranged from minimum to maximum, S_i^{rj} denotes the impedance signal at the ith frequency point ($i = 1, 2, \cdots, m$) corresponding to the jth resistor, m is the total number of the frequency points, ΔR_j is the resistance difference between r_{j+1} and r_j. In the equation, the impedance signal S can be represented by different impedance parameters including Z, θ, R, X, G and B.

Similarly, the *CCD* and *RSDR* can be expressed as following:

$$CCD_j = 1 - \sum_{i=1}^{m} \frac{\left[S_i^{r_{j+1}} - \overline{S^{r_{j+1}}} \right] \left[S_i^{r_j} - \overline{S^{r_j}} \right]}{\sigma^{r_{j+1}} \sigma^{r_j}} \tag{12}$$

$$RSDR_j = \frac{\left| CCD_{j+1} - CCD_j \right|}{\Delta R_j} \tag{13}$$

where CCD_j and $RSDR_j$ denote the cross-correlation distance and relative shape deviation rate of the impedance signals corresponding to the jth and $(j + 1)$th resistance, respectively. $\overline{S^{r_j}}$ and σ^{r_j} denote the mean and standard deviation of the impedance signal corresponding to the jth feedback resistances r_j, respectively.

In order to make the best use of the deviation characteristics of impedance signal, six kinds of impedance parameters, including Z, θ, R, X, G and B, can be combined together using the corresponding weighted accumulation to help to evaluate the impedance comparation performance and to search for the optimal feedback resistance with the minimum RADR and RSDR values.

3.2. Calibration and calculation procedure

The detailed selection procedure of the optimal feedback resistance using the proposed self-calibration method was summarized as following:

1) Data correction: Based on above analysis, for each alternative resistor, its original impedance was firstly measured by the board following the calibration procedure described in section 2.2, and then the measured impedance data were corrected one by one for all the alternative resistors following the equations (2)–(8).
2) Data arrangement: Arranging the correction data samples following the ascending order for the corresponding selected alternative feedback resistance from the minimum to maximum.
3) RADR and RSDR calculation: Calculating the RADRs and RSDRs between two corrected impedance samples corresponding to two adjacent alternative feedback resistances following the data arrangement in step 2).
4) Resistor ranking: Comparing the calculated RADRs and RSDRs, the corresponding resistors were re-arranged following the ascending order for the RADRs and RSDRs from the minimum to the maximum, respectively.
5) Weighted rating: In this study, a weighted score-based method was employed to help to evaluate the performance of the selected resistance for the impedance measurement. For example, after the RADRs and RSDRs for different kinds of impedance parameters (such as Z, θ, R, X, G or B) were calculated respectively and re-arranged following

the steps 3) and 4), a weighted score P_i of the RADR and RSDR for the ith resistance in step 4) can be calculated respectively as following:

$$P_i^{RADR} = \omega^Z P_i^z + \omega^\theta P_i^\theta + \omega^R P_i^R + \omega^X P_i^X + \omega^G P_i^G + \omega^B P_i^B \tag{14}$$

$$P_i^{RSDR} = \omega^Z P_i^z + \omega^\theta P_i^\theta + \omega^R P_i^R + \omega^X P_i^X + \omega^G P_i^G + \omega^B P_i^B \tag{15}$$

where P_i^Z denotes the ranking score of the impedance modulus |Z| corresponding to the ith alternative feedback resistance and can be determined by its ranking of the RADR and RSDR in step 4). Similarly, the P_i^θ, P_i^R, P_i^X, P_i^G, P_i^B can also be determined, respectively. ω^z, ω^θ, ω^R, ω^X, ω^G, ω^B denote the weight coefficient for the corresponding impedance parameters.

It should be noted that the weight coefficient can be selected depending on the specific situation. For example, in this study, only four impedance parameters (Z, θ, R, X) were attended to the impedance calibration and calculation, so the four weight coefficients can be set equal to $\omega_Z = \omega_\theta = \omega_R = \omega_X = 1/4$. In the practical applications, the weight coefficient for different impedance parameters can be set different depending on its importance. On the other hand, the ranking score P_i corresponding to the ith feedback resistance, can be determined depending on its ranking of the RADRs and RSDRs in step 4). In this study, the resistors with the smallest four RADRs or RSDRs were selected and ranked in ascending order, respectively. Then, the first score P corresponding to the smallest RADRs or RSDRs can be set to 1, for example, and the other P values can be set as 0.75, 0.5 and 0.25, respectively, depending on its ranking. It also should be noted that it is not a fixed value for these scores, and any value can be used as the score as long as following the descending order. The following experimental investigation will describe this operation in detail.

6) Summation for RADRs and RSDRs. After the weighted score P_i of the RADR and RSDR for the ith alternative feedback resistance, *i.e.*, the P_i^{RADR} and P_i^{RSDR}, were obtained, then a summary index $\overline{P_i}$, was calculated following the equation:

$$\overline{P}_i = \omega_{RADR} P_i^{RADR} + \omega_{RSDR} P_i^{RSDR} \tag{16}$$

where $\omega_{\rm RADR}$ and $\omega_{\rm RSDR}$ denote the weight coefficient of P_i^{RADR} and P_i^{RSDR} respectively. It is known that RADR describes the relative amplitude deviation rate, and RSDR describes the relative shape deviation rate, so the corresponding weight coefficient can be selected depending on the detailed requirement in the practical application. For example, the $\omega_{\rm RADR}$ can be set larger than the $\omega_{\rm RSDR}$ when the shape consistency was regarded as more important. In this study, the $\omega_{\rm RADR}$ and $\omega_{\rm RSDR}$ were set equally to 0.5, which means the amplitude deviation and shape deviation share the same importance for the impedance measurement.

7) Selection for the optimal resistance: Once the summary core $\overline{P_i}$ was obtained, the resistor with the first maximum summary score can usually be regarded as the optimal one.

The entire calibration and selection procedure using the proposed method is presented in Fig. 4.

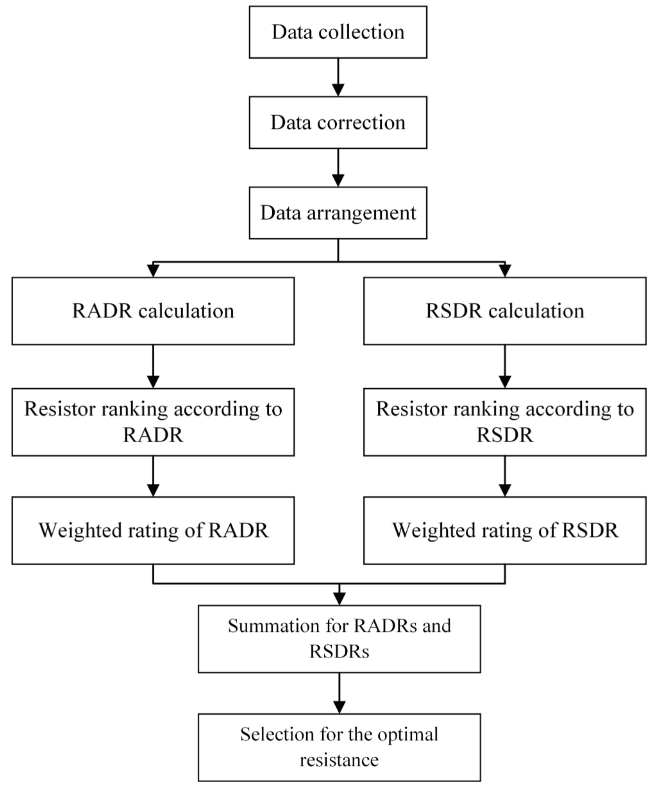


Fig. 4. Calculation and selection procedure for the optimal resistor using the self-calibration method.

4. Pretest investigation

In order to investigate the performance of the impedance measurement for the AD5933 board when employing the inappropriate feedback resistance during the calibration procedure, meanwhile, to clearly present the principle and characteristics of the proposed method for its feedback resistors selection, a pretest was conducted in this section. At first, 23 different feedback resistors with the resistance from 22 Ω to 100 kΩ (including 22 Ω, 47 Ω, 100 Ω, 220 Ω, 470 Ω, 1 kΩ, 2 kΩ, 2.2 kΩ, 3 kΩ, 4.7 kΩ, 5.1 kΩ, 7.2 kΩ, 10 kΩ, 11 kΩ, 12 kΩ, 15.1 kΩ, 20 kΩ, 22 kΩ, 24 kΩ, 27.1 kΩ, 32 kΩ, 47 kΩ, 100 kΩ) were prepared and employed to help to measure the impedance of a free PZT patch with the frequency range from 10 kHz to 100 kHz following the procedure described in Section 2.2. The parameters of the PZT patch are presented in Table 2. In the measurement procedure, the PGA was set to × 1, and the output excitation voltage was equal to 1 V.

In the pretest, the professional impedance analyzer, WK6500B (Wayne Kerr Electronic Co., UK), was employed here to provide the reference impedance signal. Finally, the measurement results were shown in Fig. 5. From the figure, it can be seen that when the selected feedback resistances used is larger than 7.2 kΩ, a gain saturation phenomenon is observed and induces a platform segment on the measured impedance signal. At the same time, with the increase of the resistance, the platform segments occur earlier and earlier. This gain saturation phenomenon can be explained by the system gain caused by the inappropriate resistance selection, and finally induces the $V_{\rm out}$ out of the ADC allowance range, and thus resulting in the inaccurate impedance measurement [32].

Fig. 6 presents the impedance measurements of the PZT patch with the frequency range from 10 kHz to 45 kHz with the feedback resistances of 22 Ω, 47 Ω, 3 kΩ, 15.1 kΩ, 20 kΩ. From the figure, it is found that the impedance signal becomes unstable when the resistance is selected too small (< 3 kΩ). At the same time, with the decrease of the resistance, the fluctuation becomes worse and worse. This phenomenon can also be attributed to the inappropriate feedback resistance selection. From the

Table 2
The parameters of the free PZT patch.

Parameters	Values	Unit
Dimension	10 × 10 × 0.5	mm
Density	7800	kg/m^3
Young's modulus	46	Gpa
Poisson's ratio	0.3	—
Dielectric loss factor	0.02	—
Mechanical loss factor	0.001	—
Piezoelectric strain coefficients d_{31}, d_{31}, d_{31},	−2.10/5.00/5.80	10^{-10} mV^{-1} or CN^{-1}
Electric permittivity ε_{11}^T, $\varepsilon_{22}^T/\varepsilon_{33}^T$	1.75/2.12	10^{-8} Fm^{-1}

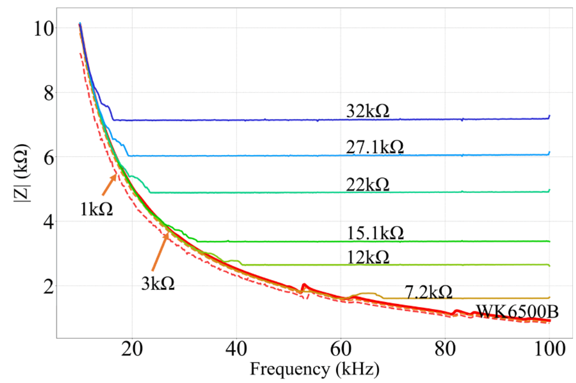


Fig. 5. Impedance measurement with different feedback resistances from 1 kΩ to 32 kΩ.

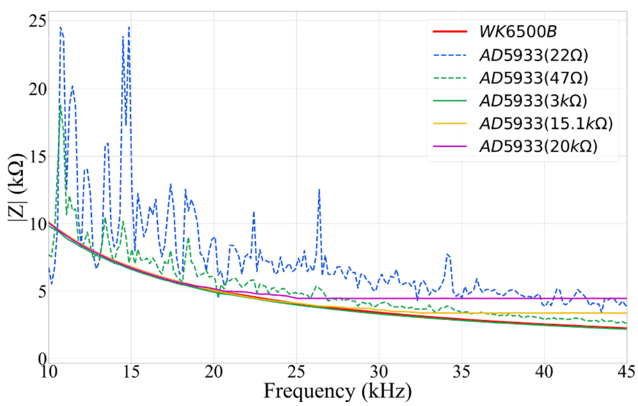


Fig. 6. Impedance measurement with different feedback resistances from 22 Ω to 20 kΩ.

Fig. 5 and Fig. 6, it is observed that the impedance measurement with the feedback resistance of 3 kΩ is closest to the reference signal, and both signals have a good consistency with each other.

In addition, the RADRs of the impedance signals measured with the help of the adjacent feedback resistance had also been calculated and plotted as shown in Fig. 7. From the figure, it is clear that the RADR presents the minimum at the position corresponding to the feedback resistance of 3 kΩ.

Based on above analysis, the impedance measurement using the AD5933 board presents several characteristics: (1) When the feedback resistance is selected larger than the optimal one, a platform segment on the measured impedance signal occurs because of the gain saturation. With the increase of the selected resistance, the platform segments occur earlier and earlier, and induce the deviation more and more obvious. (2) When the feedback resistance is selected less than the optimal one, with the decrease of the resistance, the fluctuation becomes worse and worse.

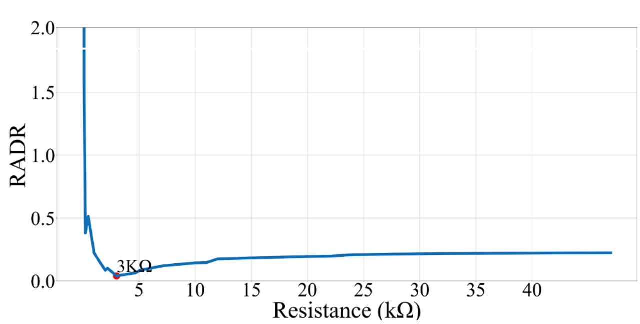


Fig. 7. RADRs corresponding to different resistance of the feedback resistors.

(3) The closer to the optimal value for the feedback resistance, the smaller of the deviation of the impedance measurement presents. The impedance measured by the board with the help of the feedback resistor with the optimal resistance presents the minimum in the RADR or RSDR index, and the impedance measurement in this situation has the best consistency with the reference one measured by the professional impedance analyzer.

Based on these characteristics, a novel self-calibration method was proposed in this study to provide the AD5933 board the potential of automatic measurement. The detailed calculation and selection process of the optimal feedback resistor can be referred to the description in section 3. With the help of this method, the resistor with the least deviation of the impedance signal can be selected and regarded as the optimal feedback resistance, and then attend to the following impedance measurement using the board.

5. Experimental investigation

In order to validate the impedance measurement performance of the proposed method, and to investigate its capability and feasibility in structural damage identification at the same time, a series of experiments were conducted in this study.

5.1. Impedance measurement performance

(1) Experimental Setup

In order to investigate the impedance measurement performance of the proposed method, three different specimens were prepared in the laboratory, including a free PZT patch, a rectangle thin steel plate and a square steel plate, as shown in Fig. 8. The detailed physical and material parameters of these specimens are presented in Table 2 and Table 3. In the experiment, the corresponding PZT patches were surface bonded on the center of the specimen with epoxy adhesive [18], and both the rectangle plate and the square plate were fixed on two wood blocks,

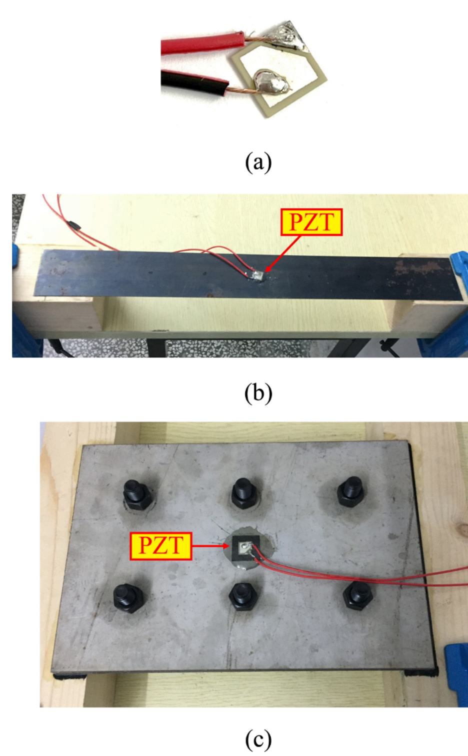


Fig. 8. Three specimens (a) free PZT patch. (b) rectangle thin steel plate. (c) square steel plate.

Table 3
The parameters of the specimens.

Specimens	Parameters	Values	Unit
Rectangle plate	Dimension	400 × 50 × 0.5	mm
	Density	7860	kg/m^3
	Young's modulus	209	Gpa
Square plate	Dimension	260 × 180 × 4	mm
	Density	7860	kg/m^3
	Young's modulus	209	Gpa

respectively.

In the experiment, 20 alternative feedback resistors were employed here with the resistance from 470 Ω to 100 kΩ, including 470 Ω, 1 kΩ, 2 kΩ, 2.2 kΩ, 3 kΩ, 3.3 kΩ, 4.7 kΩ, 5.1 kΩ, 7.2 kΩ, 10 kΩ, 11 kΩ, 12 kΩ, 15.1 kΩ, 20 kΩ, 22 kΩ, 24 kΩ, 27.1 kΩ, 32 kΩ, 47 kΩ, 100 kΩ. The AD5933 board was employed here as the impedance measurement device with the excitation voltage of 1 V and the PGA gain of × 1. In this situation, seven different test statuses were investigated, as shown in Table 4.

In this study, the subrange calibration method described in section 2.3 was also employed here as one of the calibration methods for AD5933 board for the following impedance measurement comparation. For this method, the average gain factors were obtained using the limiter average filtering method [32] at the frequency point of 10 kHz, and the finally calculated gain factors and corresponding feedback resistance were shown in Table 5.

(2) Test results and analysis

For the seven test statuses, the electrical impedance signals of the PZT patch corresponding to different specimens and investigating frequencies were measured using the AD5933 board combining with the subrange calibration method and the proposed self-calibration method, respectively, following the procedures described in section 2 and section 3. Finally, the corrected impedance signals for different test statuses were obtained and presented in Fig. 9, in which, the numerical symbol on the curve indicates record overflow. In this study, the professional impedance analyzer, WK6500B, was also employed here to provide the impedance measurement as the reference one.

It is well known that the temperature variation will affect the impedance measurement and impair the identification accuracy. Therefore, to avoid this adverse effect during the measurement, all the impedance measurement were conducted in laboratory in a short time, thus the environmental temperature was regarded as constant. In addition, an electronic thermometer was also utilized in the experiment to help to indicate the temperature variation. In this condition, the differences in the measured impedance signals can only be caused by the changes of the physical parameters of the test specimen.

From the figures, it is observed that the impedance measured using the proposed self-calibration method and the subrange calibration method both present good consistency with the reference ones (WK6500B). However, for the subrange calibration method, the record overflow phenomenon occurs commonly. For example, in Fig. 9(b), comparing with the reference impedance (red line) measured from WK6500B, the impedance measured with the subrange calibration method (blue line) presents worse consistency than the one measured with the proposed self-calibration method (green line) no matter for the characteristic frequency or the amplitude. On the other hand, the broken phenomenon was observed for the impedance curve for the subrange calibration method in the figure, in which the broken location had been marked by the numbers from 1 to 5. Following the description in section 2.3, the broken phenomenon may be induced because of the record overflow of the AD5933 board. In this situation, the discontinuity issue for the measured impedance may induce the key information missing for the to-be-measured structures, thus confuse even mislead the final evaluation for the structural health condition and service status. The similar discontinuity phenomenon was observed for all the seven comparation in Fig. 9.

Table 4
Test statuses.

Statuses	Specimen	Frequency range (kHz)
#1	Free PZT patch	10–100
#2	Free PZT patch	40–60
#3	Square steel plate	10–100
#4	Square steel plate	20–40
#5	Rectangle steel plate	10–100
#6	Rectangle steel plate	15–20
#7	Rectangle steel plate	50–60

Table 5
The calculated gain factors and resistance for the subrange calibration method.

Z_x (kΩ)	R_{FB} (kΩ)	*GF*	Z_x (kΩ)	R_{FB}	*GF*
[0, 1]	0.47	4.64×10^{-7}	[20,22]	20	1.10×10^{-8}
[1,2]	1	2.21×10^{-7}	[22,24]	22	9.97×10^{-9}
[2, 2.2]	2	1.09×10^{-7}	[24, 27.1]	24	9.14×10^{-9}
[2.2, 3]	2.2	9.97×10^{-8}	[27.1, 32]	27.1	8.13×10^{-9}
[3, 3.3]	3	7.29×10^{-8}	[32, 44]	32	6.90×10^{-9}
[3.3, 4.7]	3.3	6.62×10^{-8}	[44, 47]	44	4.99×10^{-9}
[4.7, 5.1]	4.7	4.67×10^{-8}	[47, 52.1]	47	4.65×10^{-9}
[5.1, 7.2]	5.1	4.31×10^{-8}	[52.1, 69]	52.1	4.21×10^{-9}
[7.2, 10]	7.2	3.07×10^{-8}	[69, 94]	69	3.20×10^{-9}
[10,11]	10	2.19×10^{-8}	[94, 100]	94	2.32×10^{-9}
[11,12]	11	1.96×10^{-8}	[100, 220]	100	2.19×10^{-9}
[12, 15.1]	12	1.84×10^{-8}	[220, 470]	220	9.97×10^{-10}
[15.1, 20]	15.1	1.45×10^{-8}			

In order to quantitatively evaluate the consistency and the accuracy of the impedance measurement using these two calibration methods, two indexes including the cross-correlation distance (CCD) and the mean relative error (MRE), were employed and calculated here to help to evaluate the difference between the measurement of the two methods and the reference one, according to equations (12) and (17):

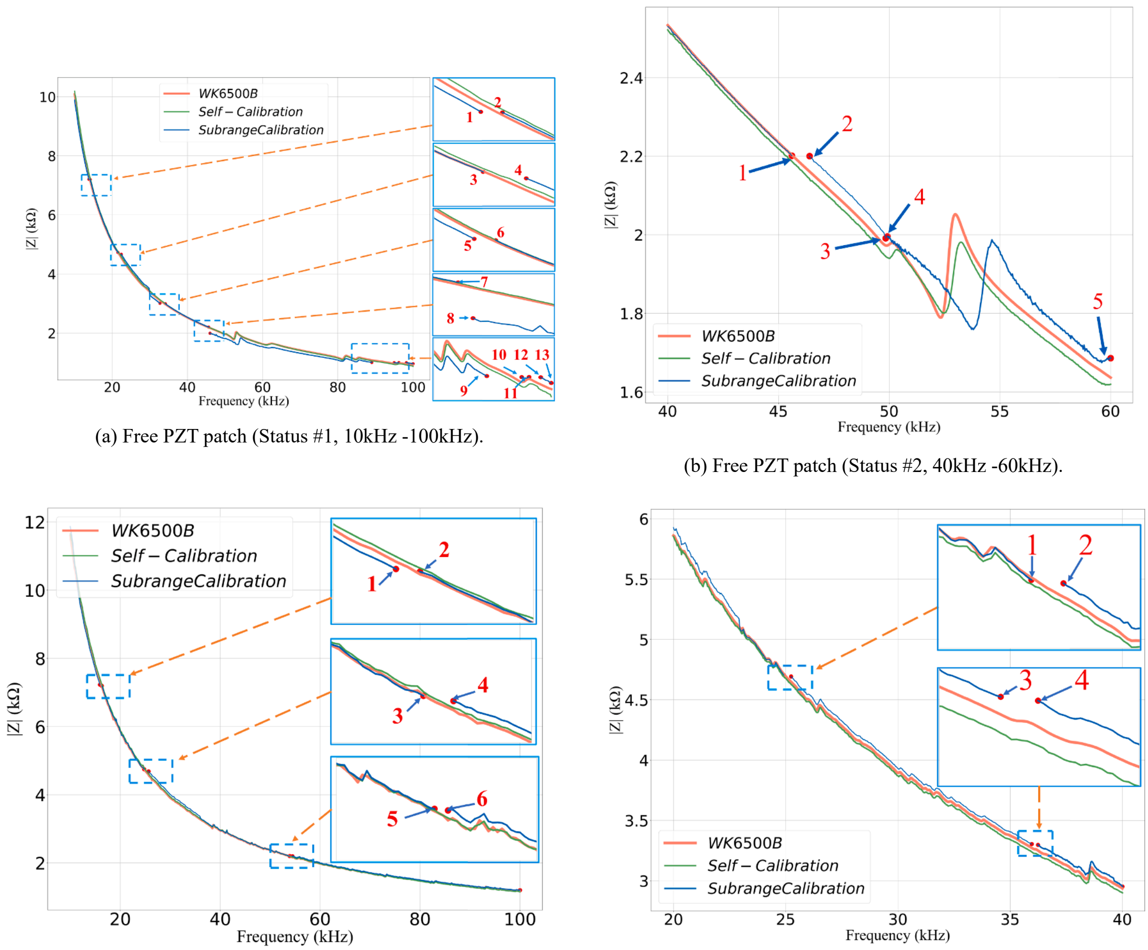
$$MRE = \frac{1}{m}\sum_{i=1}^{m}\left|\frac{s_i^1 - s_i^0}{s_i^0}\right| \tag{17}$$

where S_i^1 and S_i^0 denote the magnitude of the impedance modulus signal measured by the AD5933 board and the professional impedance analyzer, respectively, at the ith frequency point, and m is the total number of the frequency sampling points. Table 6 presents the calculation results for these two methods.

According to the results, it is clear that compared with the reference signal measured by the professional impedance measurement instrument, the AD5933 based impedance measurement using the proposed self-calibration method presents higher accuracy than the one measured using the subrange calibration method, and the maximum of MRE and CCD is less than 1.2 %. Therefore, a conclusion can be drawn that the proposed self-calibration method has the great potential to help the AD5933 board to obtain the impedance signal with enough accuracy. Meanwhile, it has also overcome the disadvantage of impedance data discontinuity for the conventional subrange calibration method.

5.2. Structural damage identification performance

In order to validate the structural damage identification feasibility of the AD5933 board combining with the proposed self-calibration method, the specimen of the square steel plate used in last section (as shown in Fig. 8(c)) was employed here to help to investigate the detection performance of bolt loosening. The experimental setup is shown in Fig. 10, in which the PZT patch was surface bonded on the center of the specimen. In the experiment, six loosening statuses were



(a) Free PZT patch (Status #1, 10kHz -100kHz).

(b) Free PZT patch (Status #2, 40kHz -60kHz).

(c) Square steel plate (Status#3, 10kHz-100kHz).

(d) Square steel plate (Status#4, 20kHz-40kHz).

Fig. 9. The impedance measurement results for test statuses #1–7.

introduced to the bolt #1 from 50 N·m (tightest status) to 0 N·m (complete loosening) with the interval of 10 N·m by using the torque wrench, meanwhile the other bolts keep the tightest status.

It is well known that the sensitivity of the impedance-based damage detection is closely related to the selected frequency range. Prior to the test, the PZT patch on the host structure had been scanned over a wide frequency range, from 10 Hz to 100 kHz (the upper limit of the AD5933 board), to select a suitable range. Finally, the frequency range, from 20 kHz to 40 kHz, was selected since several significant peaks in the impedance modulus curve were observed and may provide more obvious characteristics for the structural defect identification.

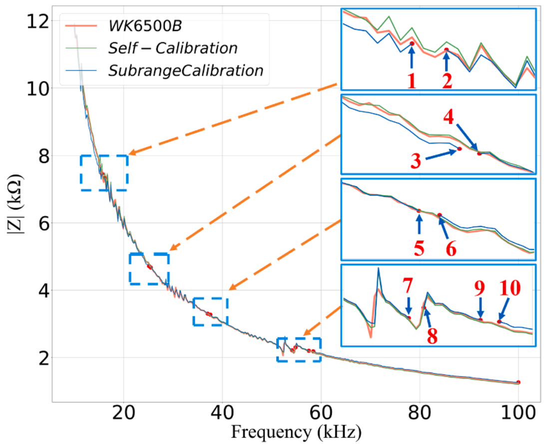
Fig. 11 presents the impedance measurement for bolt#1 of the specimen under the torque condition of 10 N·m (*i.e.*, 40 N·m loosening severity) and 50 N·m (*i.e.*, health condition as the baseline). From the figure, it is clear that the impedance signal measured by the AD5933 board has a good consistency with the one measured by the WK6500B, and this result further validates the conclusion that the AD5933 board combining with the proposed method has a great potential to be used as the impedance measurement device with enough measurement accuracy.

In this experimental study, the CCD index [33] was employed here as the damage identification index, because of its distinguished advantages of immunity to noise and sensitively to defect, thus the bolt loosening variation of the specimen can be quantificationally evaluated, as shown in Fig. 12. From the figure, it is observed that the CCD index increases with the development of the loosening severity. Besides, the identification result of the AD5933 board also presents a good consistency with the ones measured by the WK6500B. Thus, the feasibility of the AD5933 board combining with the proposed method can be successfully validated in the area of structural damage identification. Due to the space limitation, the effect of ambient environmental variation, such as the temperature, noise and so on, to the impedance results was not investigated here.

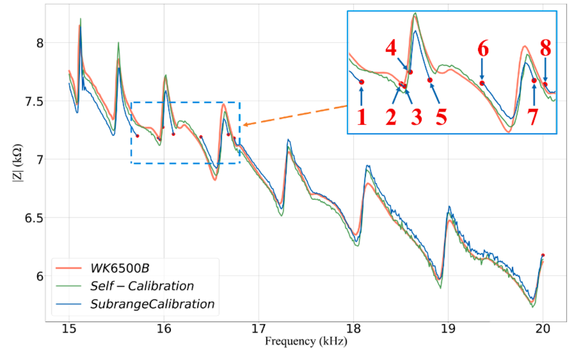
5.3. Discussion

Based on above analysis, it is found that compared with the reference impedance measured by the professional impedance measurement instrument, the impedance result measured by the AD5933 board combining with the proposed self-calibration method presents better performance for the accuracy and data continuity than the one measured by the board with the conventional subrange calibration method. In addition, its structural damage identification feasibility of the proposed method has also been validated by the experiment of bolt loosening detection in the laboratory.

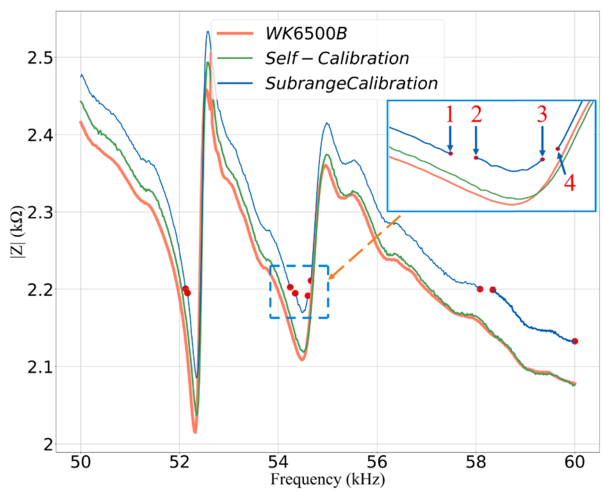
However, it should be noted that for the proposed self-calibration method, all the alternative resistors as the feedback resistance should attend to the calibration calculation at the beginning of the impedance measurement, thus a lot of computational work and time consumption will be introduced inevitably. Besides, the limitations of the AD5933 board, such as the narrow excitation frequency band, the less sampling number, and the worse measurement accuracy still exist especially when comparing with the professional impedance measurement instrument. In addition, the adverse effect on the impedance measurement due to the



(e) Rectangle steel plate (Status#5, 10kHz-100kHz).



(f) Rectangle steel plate (Status#6, 15kHz-20kHz).



(g) Rectangle steel plate (Status#7, 50kHz-60kHz).

Fig. 9. (*continued*).

Table 6
Accuracy comparation for these two impedance measurement methods.

Statuses	Self-calibration method		Subrange calibration method	
	MRE (%)	CCD (%)	MRE (%)	CCD (%)
#1	1.12	0	4.73	9.63
#2	1.14	0.30	2.14	5.93
#3	0.83	0	1.14	1.36
#4	0.72	0	0.97	2.31
#5	0.88	0.01	1.31	2.01
#6	0.56	0.45	0.98	11.12
#7	0.61	1.03	2.40	4.31

ambient environmental variation, such as the temperature, noise and so on, is another great challenge for the AD5933 board to the applied in the practical applications. For example, the temperature variation will result in the change of resonant frequencies and amplitude of the EMI signals [34], and the noise will corrupt the response signal and impair the damage diagnosis [35]. Fortunately, researchers pay a lot of attention to these issues and several distinguished methods have been proposed to eliminate or compensate these adverse effect [36–40]. In this paper, due to limited space, the temperature or noise compensation during the impedance measurement using the AD5933 board combining with the proposed calibration method was not presented here, but the related study is important and necessary, and will be further investigated in our next works.

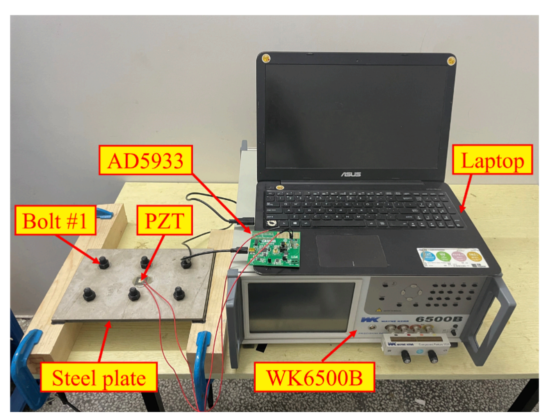


Fig. 10. Experimental setup.

In summary, with the help of the proposed method, several meaningful works had been conducted for the AD5933 board in this study to improve its feasibility and effectiveness for the impedance measurement especially when being applied in the practical applications, but more attempts still need to be done in the future works, such as embedding the related algorithms into the chip to improve the automaticity, importing

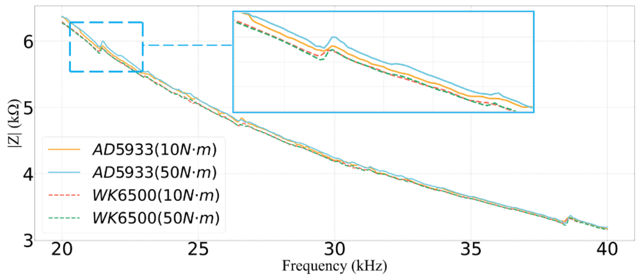


Fig. 11. Impedance measurement results for WK6500B and AD5933 board.

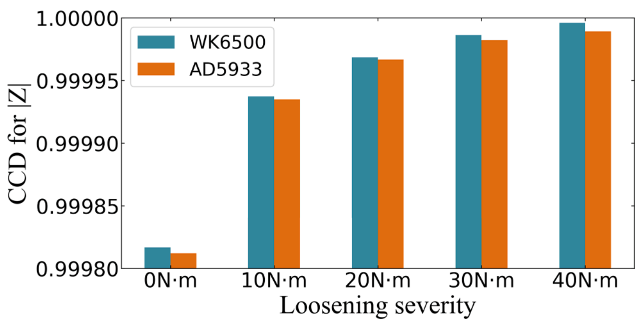


Fig. 12. CCD results of bolt #1 under different bolt torque conditions.

the wireless measurement and transmission and so on.

6. Conclusion

In order to overcome the discontinuous issue and improve the accuracy and independence for the impedance measurement by the AD5933 impedance evaluation board, a novel self-calibration method was proposed and investigated in this study. At first, the impedance calibration and measurement methodology for the board was introduced especially for the conventional subrange calibration method. Then, the principle and operation procedure of the proposed self-calibration method was presented in detail. In addition, a pretest for a free PZT patch was conducted to present the characteristics of the impedance measurement using the board especially when employing the inappropriate feedback resistance during the calibration procedure. At last, the impedance measurement performance and its capability in structural damage identification were investigated and validated by a series of experiments in the laboratory.

Finally, based on above analysis and experimental results, a conclusion can be drawn that the proposed self-calibration method in this study provides the AD5933 board: (1) a possibility to get rid of the limitation of the commercialized impedance instrument, and the ability to realize the impedance measurement independently. (2) the excellent data continuity performance in the impedance measurement comparing with the traditional calibration method. (3) the ability to measure the impedance with higher precision and better consistency with the results measured by the professional impedance analyzer. Besides, its structural damage identification ability of the proposed method combining with the board has also been successfully validated benefiting from these distinguish advantages. All these works in this study significantly improve the feasibility and efficiency of the AD5933 board for the impedance measurement, and provide the board a great potential to be applied in the areas of NDT and SHM technology especially for practical applications.

CRediT authorship contribution statement

Zhisen Tan: Conceptualization, Methodology, Writing – original draft. **Qian Feng:** Investigation, Formal analysis. **Tianjiao Ma:** Investigation, Validation. **Jingbo Zhang:** Investigation. **Yabin Liang:** Supervision, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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